

Week 6 — Model Tuning and Validation

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Objective

Optimize and validate a Random Forest classifier using cross-validation and hyperparameter tuning (Grid Search and Randomized Search), and diagnose bias-variance behavior with a learning curve.

Dataset

Pima Indians Diabetes Database — 768 patients, 8 medical predictor features (Pregnancies, Glucose, Blood Pressure, Skin Thickness, Insulin, BMI, Diabetes Pedigree Function, Age), binary outcome (diabetic / non-diabetic). Class distribution: 65.1% non-diabetic, 34.9% diabetic — a moderate imbalance.

Data Preparation

- Five columns (Glucose, BloodPressure, SkinThickness, Insulin, BMI) used 0 as an implausible placeholder for missing data — these were converted to NaN and median-imputed.
- Insulin had the most missingness (48.7% zeros), followed by SkinThickness (29.6%).
- Data split 80/20 (614 train / 154 test), stratified to preserve class balance.

Baseline Model

A default Random Forest classifier was trained first, then validated with 5-fold stratified cross-validation to check whether the single-split result was representative.

Metric	Single Split	5-Fold CV Mean
Accuracy	0.7792	0.7638 ($\pm$ 0.0214)

The close match between the single split and the CV mean indicated the baseline result was stable rather than a lucky draw.

Hyperparameter Tuning

Grid Search vs. Randomized Search

Both methods tuned `n_estimators`, `max_depth`, `min_samples_split`, and `min_samples_leaf` (Random Search additionally searched `max_features`), scoring on ROC-AUC with 5-fold cross-validation.

Method	Combinations Tested	Best CV ROC-AUC	Time (s)
Grid Search	108	0.8391	113.0
Random Search	40 (of 4,536 possible)	0.8417	74.4

Random Search matched Grid Search's cross-validation performance using roughly a third of the combinations and 34% less time — a meaningful efficiency gain at this search-space size.

Test-Set Results: Baseline vs. Tuned

Both tuned models were evaluated on the held-out test set rather than trusting the cross-validation score alone. The result was the opposite of what the CV scores predicted:

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline Random Forest	0.7792	0.7273	0.5926	0.6531	0.8192
Tuned (Grid Search)	0.7338	0.6512	0.5185	0.5773	0.8128
Tuned (Random Search)	0.7143	0.6316	0.4444	0.5217	0.8026

Every metric declined after tuning, for both search methods. The Grid Search model — the less-degraded of the two — was carried forward as the reference tuned model (`max_depth=5`, `min_samples_leaf=1`, `min_samples_split=10`, `n_estimators=100`). Its recall on the diabetic class fell most sharply, from 59.3% to 51.9%, which matters clinically: a lower recall means more diabetic patients being classified as non-diabetic.

Bias-Variance Diagnosis

A learning curve was plotted for the tuned model across increasing training set sizes. At full training size, the training accuracy was 0.8521 versus a cross-validation accuracy of 0.7721 — a gap of 0.0801. This gap indicates the model still overfits somewhat even after tuning, though tuning did constrain it relative to an unconstrained default forest.

Interpretation

- Grid Search's best cross-validation ROC-AUC (0.8391) was higher than the baseline's implicit CV performance, yet the tuned model performed worse on the untouched test set — a reminder that optimizing for CV score does not guarantee generalization, especially on a moderately small, imbalanced medical dataset (768 rows, 35% positive class).
- Random Search reached a comparable CV score to Grid Search in less time, but its test-set result was the weakest of the three models — underscoring that CV-selected hyperparameters need test-set confirmation before being trusted.
- The baseline Random Forest, run with scikit-learn defaults, remains the best-performing model on every test-set metric in this run.

Conclusion

This week's core lesson was methodological, not just technical: hyperparameter tuning that improves cross-validation score is not the same as tuning that improves real-world performance. Validating tuned candidates against a genuinely held-out test set — rather than stopping at the CV score — was what surfaced this gap. On this dataset, the untuned baseline was the better choice, and knowing that required deliberately checking rather than assuming tuning helps.